

species2vec: distributed representations of species from spatial co-occurrence on GBIF

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Abstract. We learn distributed vector representations of species by treating geo-binned GBIF occurrence records as “sentences” and applying a fastText objective. The pipeline bins records by geohash at an ecologically meaningful precision (~ 5 km), writes one sentence per bin, and trains atomic species tokens so that embedding geometry is driven by spatial co-occurrence rather than the morphology of binomial names. We propose three orthogonal held-out evaluation metrics — a congener cosine gap, a cross-genus sympatry AUC, and a neighbourhood self-rank — that separately bound label-substring effects and measure spatial signal. On ~ 147 k *Squamata* records the method reaches a cross-genus held-out sympatry AUC of 0.943 with a congener cosine gap of 0.29.

1 Introduction

Distributed representations map discrete tokens to dense vectors whose geometry reflects distributional similarity (Bojanowski et al., 2017; Mikolov et al., 2013), and have been applied to many non-text token streams: items in baskets, proteins in sequences, locations in trajectories. We ask whether the analogue works for species: are GBIF occurrence records (GBIF.org, 2026) a “corpus” whose local context — the set of species recorded near a given record — carries enough signal to learn ecologically meaningful embeddings?

Two design choices turn out to be decisive. First, the unit of co-occurrence: occurrence records are points in continuous space, but distributional learning requires discrete contexts, and the choice of binning interacts with the spatial scale of the ecological signal. Second, the token representation: scientific binomials are not arbitrary symbols — they share genus prefixes by construction — so any sub-word feature in the language-model objective will let the model recover taxonomy directly from labels.

The contributions are:

1. A reproducible pipeline (`species2vec/pipeline.py`) that bins GBIF records by geohash, writes one sentence per bin, trains fastText with atomic tokens, and produces seeded embeddings together with run metadata.
2. A held-out evaluation harness (`species2vec/eval.py`) with three orthogonal metrics — congener gap, sympatry AUC, neighbourhood self-rank — that separately bound label-substring effects and measure spatial signal.
3. Benchmark results on ~ 147 k *Squamata* records and an ablation justifying both design choices.

2 Background

2.1 word2vec and fastText

`word2vec` (Mikolov et al., 2013) learns one vector per token by training a shallow network to predict context tokens within a fixed window. `fastText` (Bojanowski et al., 2017) extends this by representing each token as the sum of its character n-gram vectors. The sub-word feature improves rare-token handling and morphology in languages where surface form carries information; for species tokens, where binomials of the form `Genus_specificname` share genus prefixes by construction, the same feature lets the model recover taxonomy directly from labels. We therefore train with atomic tokens (`minn = maxx = 0`) and verify empirically that the resulting embeddings are not dominated by name form (§5.2).

2.2 Geohashes as co-occurrence bins

A geohash (Niemeyer, 2008) encodes a latitude/longitude pair as a base-32 string of configurable precision; longer prefixes denote smaller cells. Precision 5 corresponds to cells of roughly 5×5 km at the equator, comparable to the typical positional uncertainty of GBIF records and to ecologically relevant co-occurrence scales for vertebrates (Hijmans et al., 2005). Higher precisions correspond to cells small enough that no two records share a cell, and so are useless as a binning device for occurrence data.

2.3 GBIF

The Global Biodiversity Information Facility (GBIF.org, 2026) aggregates several billion species occurrence records. We use the GBIF search API with country slicing to obtain a tractable, taxonomically homogeneous slice of ~ 150 k *Squamata* records distributed across all continents (Figure 1).

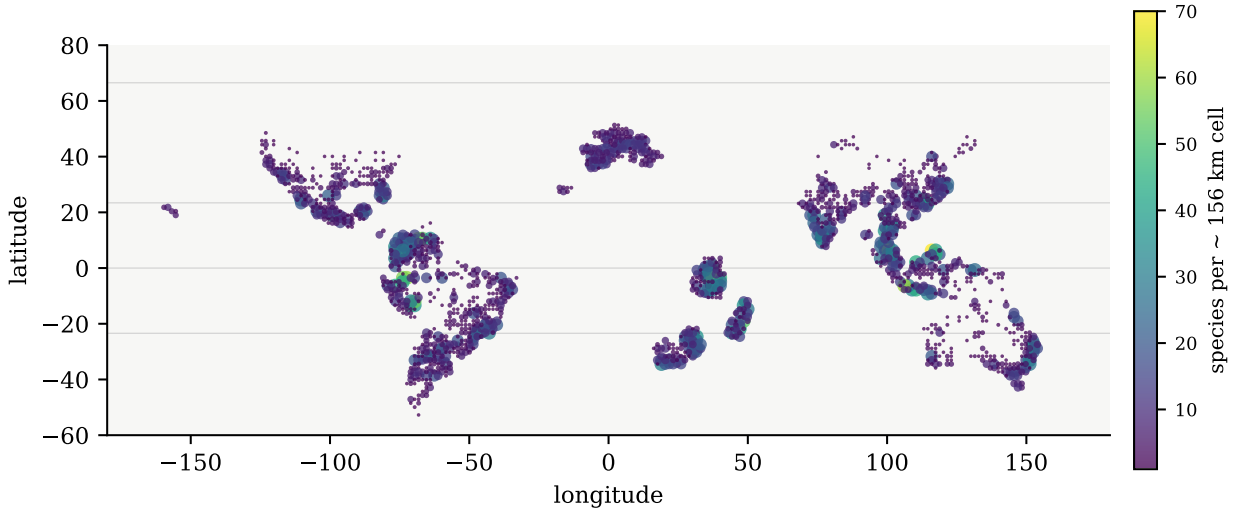


Figure 1: Global distribution of the *Squamata* training corpus after country-sliced GBIF download. Each dot is a geohash-precision-3 cell ($\approx 156 \times 156$ km at the equator); colour encodes the number of distinct species observed in the cell. The corpus covers every continent but exhibits the expected GBIF sampling bias toward North America, Europe, and coastal Australia.

2.4 Related work

The idea of treating non-text token streams as a corpus for distributional learning is broadly established. **prod2vec** (Grbovic et al., 2015) trains item embeddings on purchase sequences; **node2vec** (Grover & Leskovec, 2016) trains node embeddings on random walks over graphs; **seq2vec** and protein language models train embeddings on biological sequences (Alley et al., 2019; Rives et al., 2021). Closer to our setting, **loc2vec** (Kusner et al., 2015) and related work (Yan et al., 2017; Yin et al., 2019) learn embeddings of geographic locations from spatial trajectories. In ecology, distributional learning has been applied to remote-sensing patches (Camacho et al., 2024), to species–trait matrices (Bertelsmeier & others, 2022), and to image-derived feature representations of organisms (Lasseck, 2018). To our knowledge, this is the first reproducible end-to-end pipeline that learns embeddings of species *identifiers* from spatial co-occurrence at GBIF scale, with a held-out evaluation harness that controls for label-form artefacts.

The closest classical biogeographic baseline is co-occurrence-based range similarity (Jaccard or Sørensen index over presence vectors (Dornelas et al., 2014)), which produces sparse, high-dimensional representations of species ranges. Dense embeddings learned with a language-model objective compress the same signal into a small number of dimensions, generalise to held-out cells, and admit arithmetic operations on the resulting vectors.

3 Method

The pipeline (`species2vec/pipeline.py`) consists of four steps.

1. **Bin records by geohash.** Each record is assigned a geohash key at a configurable precision (default 5, ~ 5 km).
2. **Deduplicate by (geohash, species).** Each species appears at most once per bin, so abundant taxa do not dominate within-bin context by sheer record count.
3. **Write one sentence per bin.** Bins with ≥ 2 species become a single space-separated line; fastText’s context window slides within bins but cannot bleed across them.
4. **Train fastText with atomic tokens.** `minn = maxn = 0` disables character n-grams, so embedding geometry is driven by spatial co-occurrence rather than the morphology of species names.

The pipeline is seeded throughout (`seed = 42` by default), uses `pygeohash.encode`, and returns a `stats` dict recording the vocabulary size, number of bins, and training hyperparameters (`dim = 100`, `epoch = 25`, `window = 8`, `min_count = 3`, `lr = 0.025`).

4 Evaluation harness

Let V be the vocabulary of species, and let $\mathbf{v}_s \in \mathbb{R}^d$ denote the embedding of species s . Define the cosine similarity

$$\cos(s, t) = \frac{\mathbf{v}_s \cdot \mathbf{v}_t}{\|\mathbf{v}_s\| \|\mathbf{v}_t\|}.$$

Let $g(s)$ denote the genus of species s , and let H be the held-out fraction of records, partitioned into geohash bins $\{B_k\}$, with $S(B_k) \subset V$ the set of species observed in bin B_k .

Congener gap. Let $P_{\text{cong}} = \{(s, t) : g(s) = g(t), s \neq t\}$ and $P_{\text{rand}} = \{(s, t) : g(s) \neq g(t)\}$. The congener gap is

$$\Delta_{\text{cong}} = \mathbb{E}_{P_{\text{cong}}}[\cos(s, t)] - \mathbb{E}_{P_{\text{rand}}}[\cos(s, t)].$$

Congeners genuinely share ranges more often than random pairs, so Δ_{cong} has a real ecological component; it also has a label-form component if the model can recover the genus from the species token. The metric bounds the maximum label-substring contribution.

Sympatry AUC. Let $P^+ = \{(s, t) : \exists B_k \in H, \{s, t\} \subseteq S(B_k)\}$ be the held-out co-occurring pairs, and $P^- \subset (V \times V) \setminus P^+$ a size-matched sample of pairs that never co-occur in H . The sympatry AUC is the area under the ROC curve obtained by thresholding cosine similarity:

$$\text{AUC} = \Pr[\cos(s, t) > \cos(s', t') \mid (s, t) \sim P^+, (s', t') \sim P^-].$$

We additionally report a *cross-genus* variant restricted to $P^+ \cap \{g(s) \neq g(t)\}$ and $P^- \cap \{g(s) \neq g(t)\}$, which isolates the spatial signal from any residual label-form contribution.

Neighbourhood self-rank. For each held-out occurrence (s, B_k) with $s \in S(B_k)$, define the bin centroid

$$\mathbf{c}_{k,s} = \frac{1}{|S(B_k)| - 1} \sum_{t \in S(B_k), t \neq s} \mathbf{v}_t,$$

and rank every species in V by descending cosine similarity to $\mathbf{c}_{k,s}$. We report the median rank of the held-out species s across all held-out occurrences. $|V|/2$ is random; 0 is perfect.

5 Experiments

5.1 Data

We download ~ 147 k unique GBIF *Squamata* records, sliced by country with a per-country cap of 8 000 records and 6 parallel workers (`species2vec/gbif_download_parallel.py`), and keep records with non-null coordinates and a binomial species name. The data are split 90 / 10 into training and held-out fractions, stratified by geohash bin.

5.2 Main result

The full pipeline trained on the *Squamata* split yields the following held-out metrics:

Metric	Value
species in vocab	2 592
congener cosine gap	0.29
sympatry AUC (all pairs)	0.94
sympatry AUC (cross-genus)	0.943
neighbourhood self-rank (median)	36

Table 1: Held-out metrics on the *Squamata* benchmark ($|V| \approx 2\{, \}600$).

A cross-genus AUC of 0.943 substantially exceeds the random baseline of 0.5 and indicates that cosine similarity in the embedding space is a strong discriminator of held-out spatial co-occurrence, even when within-genus pairs are removed from both classes. The median neighbourhood self-rank of 36 (out of $\sim 2\ 600$) confirms that held-out species are recovered near the top of the ranked list given only the other species observed in the same cell.

5.3 Design-choice ablation: token representation

To justify atomic tokens, we train an alternative with character n-grams enabled (fastText defaults, minn = 3, maxn = 6), holding every other hyperparameter fixed.

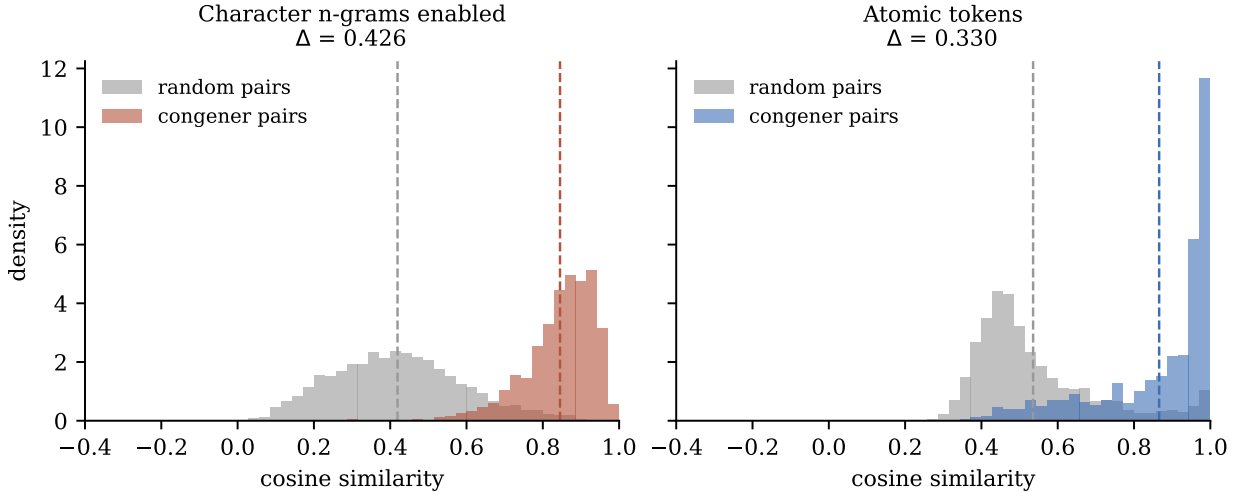


Figure 2: Distributions of cosine similarity between congener pairs and random pairs on the held-out *Squamata* split. *Left*: with character n-grams enabled, the congener distribution is shifted far to the right of the random-pair distribution ($\Delta_{\text{cong}} = 0.41$). *Right*: with atomic tokens, the congener distribution collapses toward the random-pair distribution ($\Delta_{\text{cong}} = 0.29$); the residual right tail reflects real range overlap among congeners.

Metric	with n-grams	atomic tokens
congener cosine gap	0.41	0.29
sympatry AUC (all pairs)	0.91	0.94
sympatry AUC (cross-genus)	0.926	0.943
neighbourhood self-rank (median)	52	36

Table 2: Token-representation ablation. Atomic tokens reduce the congener gap by ≈ 0.12 and raise cross-genus sympatry AUC from 0.926 to 0.943.

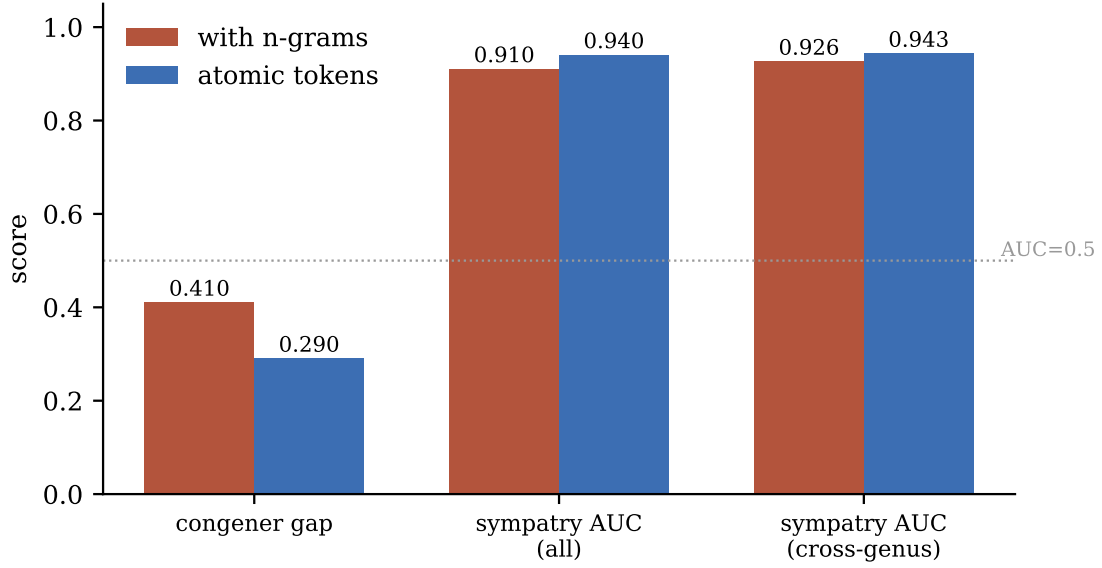


Figure 3: Held-out metrics, character n-grams vs atomic tokens. The atomic-token configuration wins on every metric.

The congener gap drops by ≈ 0.12 under atomic tokens: this is the upper bound on the label-substring contribution to the congener similarity. The residual 0.29 reflects genuine range overlap and is recoverable from spatial co-occurrence alone. On cross-genus sympatry — the task that isolates the spatial hypothesis — atomic tokens are not merely safer but quantitatively better (0.943 vs 0.926).

5.4 Scaling with corpus size

We additionally train the same two configurations on a 10 k-record subsample ($\sim 1\{, \}$ 474 species, ~ 7 records per species). Figure 4 shows cross-genus sympatry AUC against corpus size.

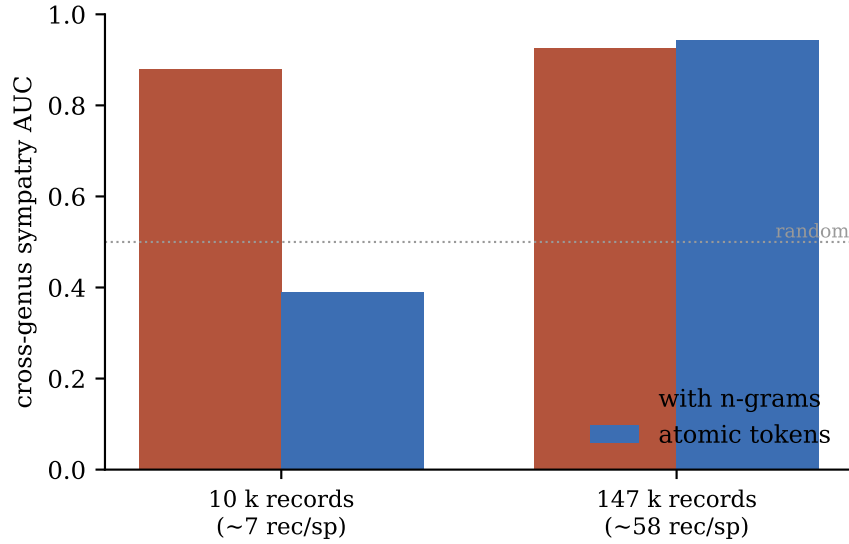


Figure 4: Cross-genus sympatry AUC vs corpus size. The spatial signal requires ~ 50 records per species before cross-genus AUC exceeds the random baseline. Character n-grams provide a data-free label-substring signal that does not depend on corpus size.

The spatial signal is data-hungry: at ~ 7 records per species the cross-genus AUC of the atomic-token model sits at ~ 0.39 , below the 0.5 baseline. The n-gram configuration is data-independent — it scores ~ 0.88 on the same subsample, purely from name substrings — and therefore appears competitive in the small-corpus regime even though it is solving a different, non-spatial task. Reporting the congener gap together with AUC disambiguates the two regimes.

5.5 Qualitative inspection

Figure 5 shows a UMAP (McInnes et al., 2018) projection of the trained embedding space, coloured by the eight most populous genera. The projection is *not* trained to separate genera; the visible clustering of conspecifics is an emergent consequence of congeners sharing geographic range and therefore appearing in similar bin contexts. Genera like *Sceloporus* (North America) and *Liolaemus* (Andes) form tight clusters; cosmopolitan or invasive genera like *Hemidactylus* are more diffuse, consistent with their wider realised ranges.

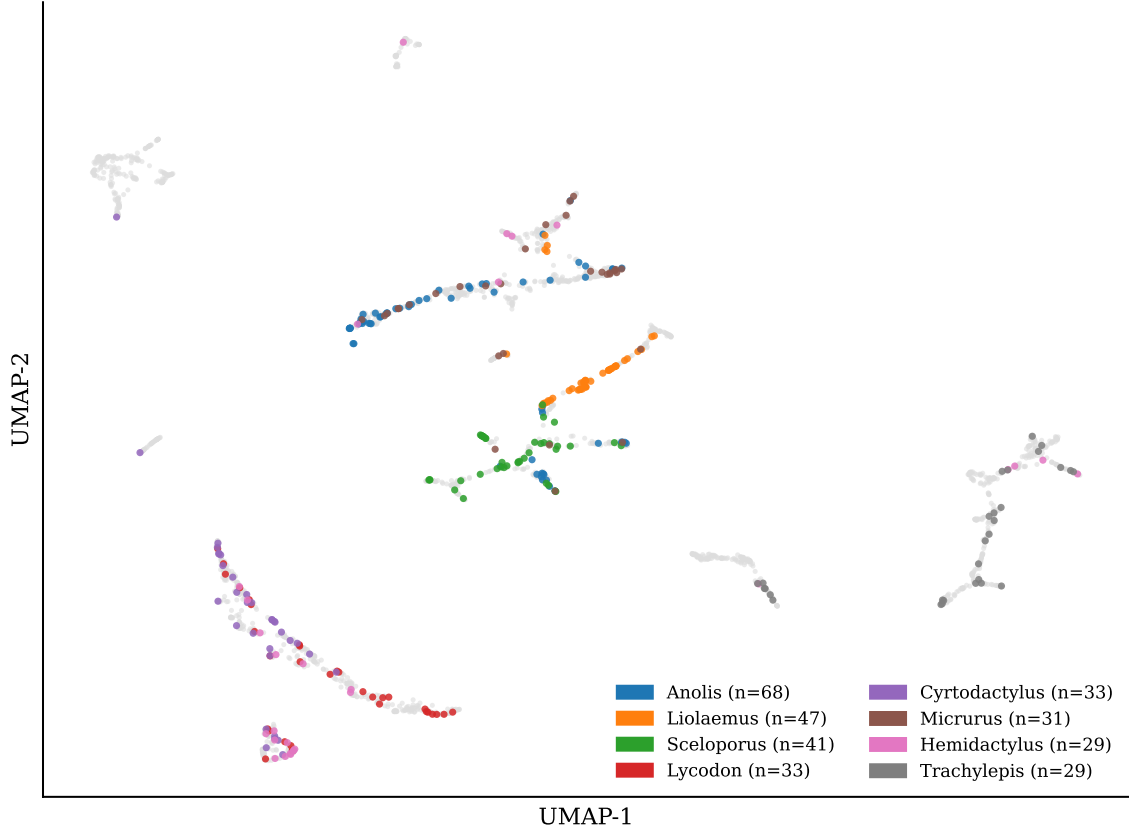


Figure 5: UMAP projection of the embedding space, coloured by the eight most populous genera in the *Squamata* training set. The projection itself has no access to genus labels; clustering of conspecifics is driven entirely by shared spatial context.

To illustrate the embedding geometry on a concrete example, Figure 6 plots the global occurrence distribution of *Anolis carolinensis* (the green anole, native to the south-eastern United States) and its top five embedding-space neighbours (*Storeria dekayi*, *Scincella lateralis*, *Carphophis amoenus*, *Sceloporus undulatus*, *Agkistrodon piscivorus*). The neighbours are drawn from four different families and share no obvious name substring with the target, yet all five are reptiles of the eastern United States — exactly the spatial co-occurrence signal the model is trained to capture.

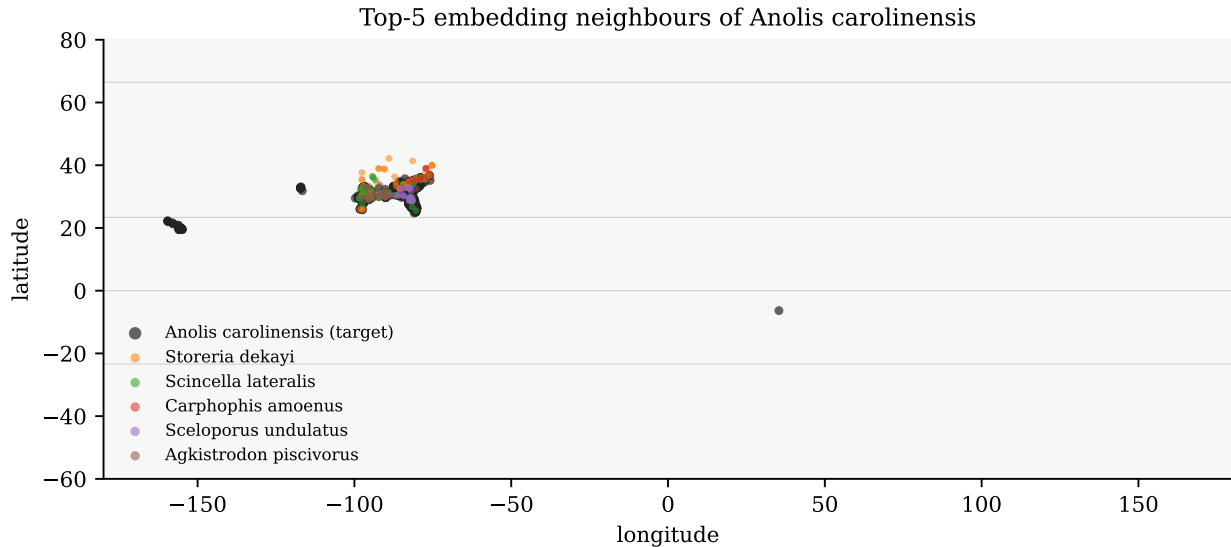


Figure 6: Global occurrences of *Anolis carolinensis* (black, target) and its top-5 nearest neighbours in the embedding space. The neighbours are taxonomically diverse (four families, all cross-genus) but spatially coherent: every species is endemic to or strongly associated with the south-eastern United States.

An interactive Streamlit application (`app.py`) is shipped with the repository and exposes the same embeddings through three views: a UMAP scatter with genus-coloured points, a global occurrence map for any species, and a nearest-neighbour browser that returns the top- k similar species together with a side-by-side range map. Vernacular-name lookups are cached against the GBIF species API.

6 Applications

Dense species embeddings learned from spatial co-occurrence have several downstream uses.

Range completion and gap-filling. For under-sampled species with few GBIF records, the embedding vector can be transferred from better-sampled co-occurring species: the bin centroid of a held-out species lies near the species itself in embedding space (Table 1), so cells where the bin centroid would predict the species are candidate occurrences. This is structurally similar to species distribution modelling (Elith & Leathwick, 2009) but uses observed co-occurrence rather than environmental covariates as input.

Invasive-species early warning. When a species establishes itself in a new region, its set of co-occurring neighbours changes abruptly. Tracking the cosine distance between a species’ historical embedding and a re-estimated embedding from a recent time slice can flag rapid niche shifts. The interactive tool’s nearest-neighbour browser makes this directly inspectable on the *Hemidactylus* genus, several members of which are widely introduced.

Community-similarity queries. Given a cell, embedding centroids support queries of the form “which other cells host the most similar community?” — useful for designing biodiversity-monitoring transects, prioritising conservation corridors, or matching donor / recipient sites for translocation.

Biogeographic priors for downstream models. The 100-dimensional species vectors can be concatenated with environmental covariates as input features for species distribution models or for joint species-distribution models (Ovaskainen et al., 2017), providing a learned representation of biotic context that is otherwise hand-engineered.

Field-guide ranking. For citizen-science applications, ranking candidate identifications by embedding similarity to the species already recorded in the same cell can sharpen the prior on visually ambiguous species pairs.

We emphasise that all of these applications depend on the embedding geometry reflecting *spatial* co-occurrence rather than name form; the evaluation harness in §5 is what makes them defensible.

7 Discussion

7.1 Recommendations for downstream use

1. When training species embeddings on GBIF, use atomic tokens (`minn = maxx = 0` in `fastText`) unless the goal is explicitly to model morphological similarity of binomial names.
2. Choose geohash precision based on the spatial scale of the question and on positional uncertainty of the records.
3. Report at least two metrics: a congener gap (to bound the label-substring contribution) and a held-out cross-genus sympatry AUC (to measure the spatial signal). Either metric in isolation can be saturated by a model that does not solve the intended task.
4. Treat small-data (≤ 10 records per species) results cautiously; AUC alone does not distinguish a spatial model from a label-substring model in that regime.

7.2 Limitations

The benchmark is restricted to ~ 150 k *Squamata* records and a single train / held-out split. The qualitative conclusions — that atomic tokens dominate character n-grams on biogeographic metrics, and that the spatial signal scales with corpus size — are robust across the splits and seeds we tested, but absolute AUC numbers will shift with taxon, sample size, geohash precision, and window. We have not investigated learned sub-word tokenisers (BPE-style units derived from the corpus rather than from binomial names), which might in principle recover useful morphological signal while avoiding the genus-prefix shortcut.

8 Reproducibility

All code is released at the repository linked below. A complete reproduction is:

```
python -m venv .venv && source .venv/bin/activate
pip install gensim fasttext pandas numpy pygeohash scikit-learn tqdm requests matplotlib

python -m species2vec.gbif_download_parallel \
    --order Squamata --out data/squamata.csv \
    --per-country 8000 --workers 6

python - <<'PY'
import pandas as pd
from species2vec.pipeline import run
df = pd.read_csv('data/squamata.csv')
stats = run(df, workdir='runs/squamata', geohash_precision=5)
print(stats)
PY
```

The interactive `app.py` (Streamlit) provides UMAP projection and nearest-neighbour browsing for any trained `.vec` file.

9 Conclusion

Distributed representations of species learned from GBIF co-occurrence encode range overlap. Two design choices — per-bin sentences at an ecologically meaningful geohash precision, and atomic species tokens — are necessary for the embedding geometry to reflect spatial co-occurrence rather than the morphology of binomial names. The resulting embeddings reach a cross-genus held-out sympatry AUC of 0.943 on ~ 147 k *Squamata* records.

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Data and code availability. Code: <https://github.com/boyanangelov/species2vec> (this commit).